Fiscal Year:	FY 2024	**Task Last Updated:**	FY 01/02/2025
PI Name:	Funke, Michael M.D., Ph.D.		
Project Title:	High-Density Dry-Electrode EEG in LEO Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	michael.e.funke@uth.tmc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	7135007155
Organization Name:	The University of Texas Health Science Center at Houston		
PI Address 1:	6410 Fannin St		
PI Address 2:	Ste 732		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Catalyst
Start Date:	07/01/2023	**End Date:**	06/30/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fiedler, Patrique Ph.D. (Technical University of Ilmenau) Maestu, Fernando Ph.D. (Compultense University of Madrid)		
Grant/Contract No.:	NNX16AO69A-CAT0004		
Performance Goal No.:			
Performance Goal Text:			
	Aim 1: Re-analysis of previous existing data 1.A. Comparison of electroencephalogram (EEG) derived functional connectivity metrics with 128 conventional and dry electrode recordings in 30 control subjects. To assure a high signal quality and feasibility for the inflight recordings, we will evaluate wet and dry electrode recordings obtained from 30 control participants, already available in a de-identified database. Participants underwent two EEG resting state recordings with dry and wet electrodes. The aim is to use this existing database from our collaborators at the University of Ilmenau, and analyze functional connectivity (phase locking value) in the sensor and source space in each of the modalities (dry/wet electrodes) and then test the correlation between them. We will test which links and nodes could be more affected by changing the electrode type.		

Task Description:	1.B. Comparison of EEG derived functional connectivity metrics of head down bedrest EEG. We will obtain already existing and completely de-identified bed rest EEG data (Brauns et al., 2021). We will test whether the reduction of the alpha band found in our International Space Station (ISS) EEG study (Cheron et al., 2014) is similar to the one found in Brauns et al study. This will help us to understand similarities and dissimilarities between these different conditions, and to design future experiments considering microgravity as a key factor. Aim 2: Development of requirements and specifications for 128 channel dry-electrode system and initial flight experiments 2.A. Requirements and specifications for 128 channel dry-electrode EEG system The EEG hardware and recording software needs to be evaluated and adjusted to meet the requirements for (1) hardware flight certification, and for (2) the science of the experiment. Hardware has to be capable to pass Space Exploration Technologies Corp (SpaceX) testing and certification, and durable and reliable enough for recording at different facilities for pre- and post-flight EEG assessments. Recording software needs to be easy to use for non-expert users and able to record automatically data quality measures, and allowing for some basic assessment on the ground during a flight mission. 2.B. Conceptual development of three initial EEG flight experiments Example: 1st opportunity with 14 days mission. There are two major goals for that initial opportunity using a high density dry-electrode EEG system. (1) Proof of principle: • Crew is able to successfully execute all steps needed to record EEG; • Crew records all or most data sets as per the experimental design/plan; • Data quality acceptable or lower/better as per experiment design; • Successful inflight assessment of EEG data quality. (2) EEG data collection, including inflight, with the goal to confirm the findings from a prior ISS experiment (NEUROSPAT) for the pre- and inflight data condition.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1) We demonstrated in two independent studies comparing conventional electrodes vs. dry-electrodes. Both studies underline the usability and applicability of dry electrode technology, even in a worst-case scenario, including applications in spaceflight. 2) We described in a comparison of spaceflight EEG data and a head down bedrest (HDBR) analog EEG data, that those are dissimilar. Both demonstrated different phenomena. HDBR is not a good model for EEG in space. In addition, for functional connectivity EEG analysis, a minimum of 64 electrodes/sensors is needed. 3) We demonstrated the 3rd generation of dry-electrode EEG electrodes that adapts to the head shape of the subject, and provided the requirements and specification for such a spaceflight EEG system. There is added value in the multiple use case scenarios that we developed as part of that report. 4) We developed three EEG experiment templates for acute, subacute, and long-term adaption of adaption and recovery of neurophysiological brain activity. Each experiment addresses certain knowledge gaps and risks of human spaceflight in low-Earth orbit (LEO) and/or exploration class missions.
Bibliography Type:	Description: (Last Updated: 12/09/2025)
Articles in Peer-reviewed Journals	Quivira-Lopesino A, Sevilla-García M, Cuesta P, Pusil S, Bruña R, Fiedler P, Cebolla AM, Cheron G, Funke M, Maestu F. "Changes of EEG beta band power and functional connectivity during spaceflight: A retrospective study." Sci Rep. 2025 Apr 18;15(1):13399. https://doi.org/10.1038/s41598-025-96897-5 ; PMID: 40251277 PMCID: PMC12008298 , Apr-2025
Articles in Peer-reviewed Journals	Sevilla-García M, Quivira-Lopesino A, Cuesta P, Pusil S, Bruña R, Fiedler P, Maestu F, Cebolla AM, Cheron G, Brauns K, Stahn AC, Funke ME. "Brain power comparison between microgravity and head-down tilt bedrest: An electroencephalography approach." Sci Rep. 2025 Nov 27;15:42314. https://doi.org/10.1038/s41598-025-26291-8 ; PMID: 41309707; PMCID: PMC12660935 , Nov-2025